J. Connor Sample

Harrisonburg, VA tabulate.tech connor-sample TabulateJarl8

Professional Summary

Computer Science student specializing in systems programming (Rust/C) and automation. Seeking roles in OS development, networking, or backend engineering. Backed by 12+ years of programming experience, including Linux administration, open-source contributions, and low level projects.

Technical Skills

Languages: Rust, Python, C, Java, TypeScript, Bash, HTML/CSS, SCSS, Ruby, x86 Assembly
Technologies: Git, Linux, Docker, CI/CD, Flask, Bootstrap, Qt, LaTeX, Vite, Wireshark, Vue.js, Postgres, Kafka
Concepts: Operating Systems, Networking, Data Structures, Systems Programming, FFI, Scrum, Unit Testing

Experience

Software Engineering Intern *Sterling, VA*
CACI International Inc *May 2026 – Aug 2026*

- Delivered a highly requested feature to 100k+ users using Spring Boot, PostgreSQL, TypeScript, Kafka, and Vue.js.
- Learned how to interact with distributed development environments using Kubernetes on ArgoCD.
- Participated in daily standups, sprint retros, and iterative collaboration with the team ensuring a high-quality product.

IT Assistant *Harrisonburg, VA*
James Madison University College of Science and Mathematics *Jan 2024 – Present*

- Orchestrate network-based management and imaging for 350+ desktops across campus using FOG.
- Architected and deployed a containerized Flask/React application (Docker) on a remote server to streamline workflows.
- Troubleshoot networking issues and document server maintenance procedures for the department knowledge base.

Software Developer Intern *Palmyra, VA*
Fluvanna County Public Schools IT *Aug 2022 - Mar 2023*

- Engineered a bus pass generation system using Qt and SQL, replacing legacy software used by 3 schools in the district.

Projects

Rosa (Operating System) *gitlab.com/TabulateJarl8/rosa*

- Built a cooperatively multitasking OS in Rust from scratch, featuring custom device drivers for PCI, IDE/ATA, and networking, alongside an EXT2 file system implementation.
- Navigated the ACPI specification through 50+ hours of analysis to architect the OS power management subsystem.

Velour *github.com/TabulateJarl8/velour*

- Built strongly-typed, extensible TS/Vue Fedora setup script generator with a BFS-based plugin dependency resolver.
- Maintained 100% test coverage and developed a custom fuzzer to validate generated bash permutations.

devexec *github.com/TabulateJarl8/devexec*

- Developed a Rust kernel module enabling execution of buffered writes via anonymous shared memory files.
- Used unsafe kernel interfaces to execute code directly from memory, bypassing standard security.

Personal Homelab

- Administered a bare-metal Debian server, implementing a ZFS RAIDz1 NAS and Docker container management.
- Securely exposed web services to the internet using Cloudflare Tunnel as a zero-trust reverse proxy.
- Achieved a 99.57% service uptime.

Education

James Madison University — Harrisonburg, VA *Aug 2023 - May 2027*
BS in Computer Science — Major GPA: 4.0/4.0 — Cum. GPA: 3.930/4.0

- **Courses:** Operating Systems, Computer Systems I/II, Computer Arch., Software Engineering, Data Structures & Algorithms, Discrete Math I/II, Adv. Networking, Adv. Algorithms, Graphics, Navigation Systems.

Certifications & Organizations

Linux Foundation Certified Systems Administrator (LFCS) *Aug 2025*

- The Linux Foundation • ID: LF-3x36tjsqkk

JMU Unix Users Club — President *Jan 2024 – Present*

- Present technical workshops on Linux administration and open-source technologies to diverse audiences.
- Manage club infrastructure and assist in organizing outreach events.

Upsilon Pi Epsilon (ΥΠΕ) — Member *Apr 2026 – Present*

- Inducted into international computing honor society, restricted to the top 11% of CS majors in 2026.

JMU Marching Royal Dukes — Member *Aug 2023 - Present*